DREAM CLOUD LAB

DREAM Brain: Recursive Science Improvement

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Abstract. AI can propose proteins faster than laboratories can establish what they do. Closing this gap requires a system that connects scientific objectives to executable experiments, interprets measurements in context, and carries knowledge across design cycles. We present the architecture of **DREAM Brain**, the intelligence layer being developed for the DREAM Cloud Lab, a programmable protein-engineering facility. The architecture couples evidence-grounded design, cell-free experimentation and laboratory automation with a shared scientific memory. Its organizing unit is a versioned experimental cycle that links each candidate to its supporting evidence, rejected alternatives, prospective predictions, execution provenance and subsequent interpretation. This structure supports two complementary goals: improving proteins within a campaign and improving reusable models across campaigns. We describe the interfaces needed to connect scientific planning to instrument drivers, manipulation skills and a state-aware digital twin. A human-designed RamR pilot illustrates why changes in fluorescence must be separated from validated sensing; the supplied summary data support assay qualification as the next decision, rather than a claim of improved function. Lanmodulin selectivity and PduA assembly provide two prospective tests of broader applicability. We define an evaluation program that separates scientific decision quality, execution reliability and learning benefit. DREAM Brain frames the cloud laboratory as infrastructure for producing both functional proteins and the evidence needed to improve their design.

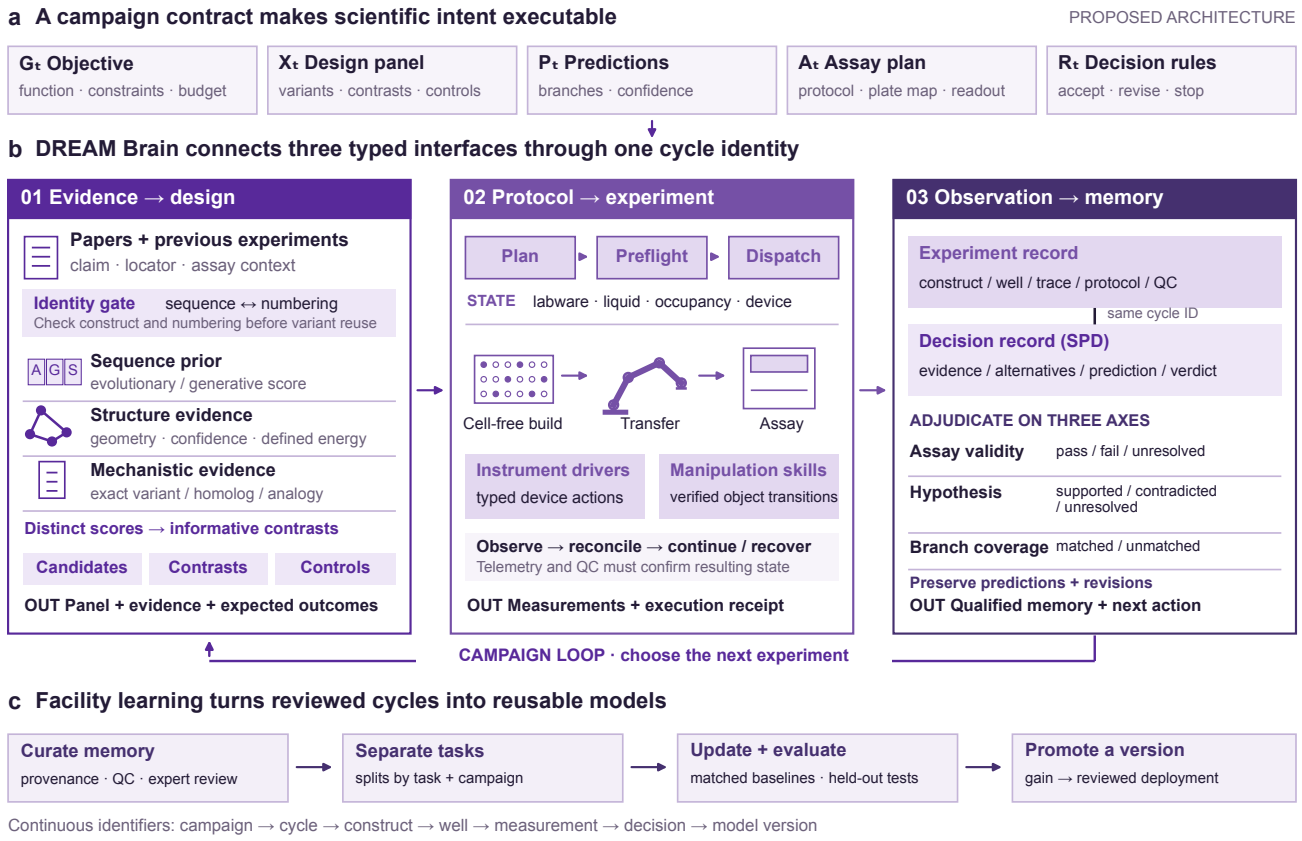


Figure 1. A learning cloud laboratory with explicit interfaces. (a) A campaign contract links objectives, candidates, predictions, assay plans and decision rules. (b) Evidence-grounded design produces a panel and expected outcomes; execution produces measurements and a state-reconciled receipt; scientific memory joins the experiment and decision records. The campaign loop uses the resulting interpretation to choose the next experiment. (c) Facility learning requires curation, task-specific partitions, independent evaluation and reviewed deployment. This is a proposed architecture, not a claim of end-to-end deployment.

1 Introduction

Protein engineering increasingly begins with computational proposals. Methods such as ProteinMPNN and RFdiffusion have expanded the range of sequences and structures that can be designed computationally [1, 2]. Yet a promising sequence is only the beginning of a functional claim. The protein must be produced, measured under relevant conditions and compared with appropriate controls. When an experiment is ambiguous, the next useful step may be to repair the assay, test an assumption or separate competing mechanisms. Increasing the number of generated candidates does not resolve these decisions. Directed evolution and machine-learning-guided protein engineering provide complementary foundations for connecting sequence proposals to experimentally measured function [13, 14, 15, 16, 17]. Deep mutational scanning further illustrates the value of systematic sequence–function measurements and the importance of interpreting variants in their experimental context [18, 19].

The opportunity for a cloud laboratory is therefore broader than increasing throughput. A shared facility can connect the design, build, test and learn stages through consistent identities, protocols and evidence. It can make advanced experimental capabilities accessible to users who do not operate the instruments, while preserving enough context for their results to inform future campaigns. Robot-scientist studies, cloud experimentation and biofoundry initiatives provide precedents for connecting computational decisions to shared experimental infrastructure [20, 21, 22, 23, 24]. Self-driving laboratory studies extend this connection to iterative experimental selection and execution [25, 26, 27, 28]. The central systems question is how to make this connection reliable: *what must travel with a proposed experiment so that it can be executed, interpreted and learned from?*

DREAM—the AI-Driven, Rapid, Experimental Automation Machine—is being developed within Northwestern University's Center for Synthetic Biology as an NSF-supported national user facility for protein engineering [3, 4]. Its mission combines AI-guided design, automated cell-free protein synthesis, functional characterization and a shared protein-function knowledge base. The publicly announced four-year program targets more than 300,000 characterized proteins and up to 30 million data points; these are facility goals, not results of this manuscript [5]. This scale motivates an intelligence layer that works across scientific objectives and experimental modalities.

We propose **DREAM Brain** as that layer. Its role is to maintain continuity between a user's scientific objective, the evidence used to choose an experiment, the physical process used to execute it and the interpretation that determines what happens next. Scientific memory is central to this architecture, but it is useful because it connects decisions to experiments and models. A record becomes a learning asset only when its evidence, temporal ordering and measurement validity are recoverable.

This framing builds on complementary advances. Coscientist demonstrated the integration of language models, documentation and automated chemistry equipment [6]. FAIR principles and provenance standards establish foundations for findable, reusable data and explicit relationships among entities, activities and agents [7, 8]. DREAM Brain extends these ideas into a campaign-level design: prospective decisions, physical provenance and assay-qualified outcomes are linked through one cycle identity. We do not claim that existing notebooks or provenance standards cannot express scientific reasoning. The proposed contribution is a concrete interface and evaluation strategy for using such information throughout a programmable protein-engineering laboratory.

Three contributions organize the paper. First, we define an architecture that connects evidence and design, physical execution and scientific memory. Second, we specify Scientific Process Data (SPD), a representation of prospective choices and subsequent interpretations that complements measurement records. Third, we use a preliminary biosensor case and two prospective campaigns to derive an evaluation program for a learning facility.

Evidence category	Available in this manuscript	What it supports
System design	Architecture, interfaces and SPD specification	A proposed integration and testable requirements
Experimental pilot	Four RamR summary profiles from a human-run cycle	An illustrative interpretation and decision case
Simulation asset	Frames from a laboratory digital-twin video	A visual workcell representation
Prospective work	Lanmodulin, PduA and comparative evaluation	A defined validation program

Table 1. Scope of the present evidence. End-to-end autonomous execution, improved protein function and benefits from process-data training remain to be demonstrated.

2 DREAM Brain: Recursive Science Improvement

DREAM Brain is conceived as a versioned scientific control layer connecting a facility's mission to experiments and their accumulated evidence. Its proposed architecture has three coupled loops (Figure 2). The **scientific design–build–test–learn (DBTL) loop** determines the next experiment. The **model-learning loop** uses qualified experimental records to improve reusable predictive and design models. The **meta-RSI loop**, where RSI denotes recursive science improvement, evaluates changes to the planning and learning process itself. These loops operate on different objects—experiments, models and

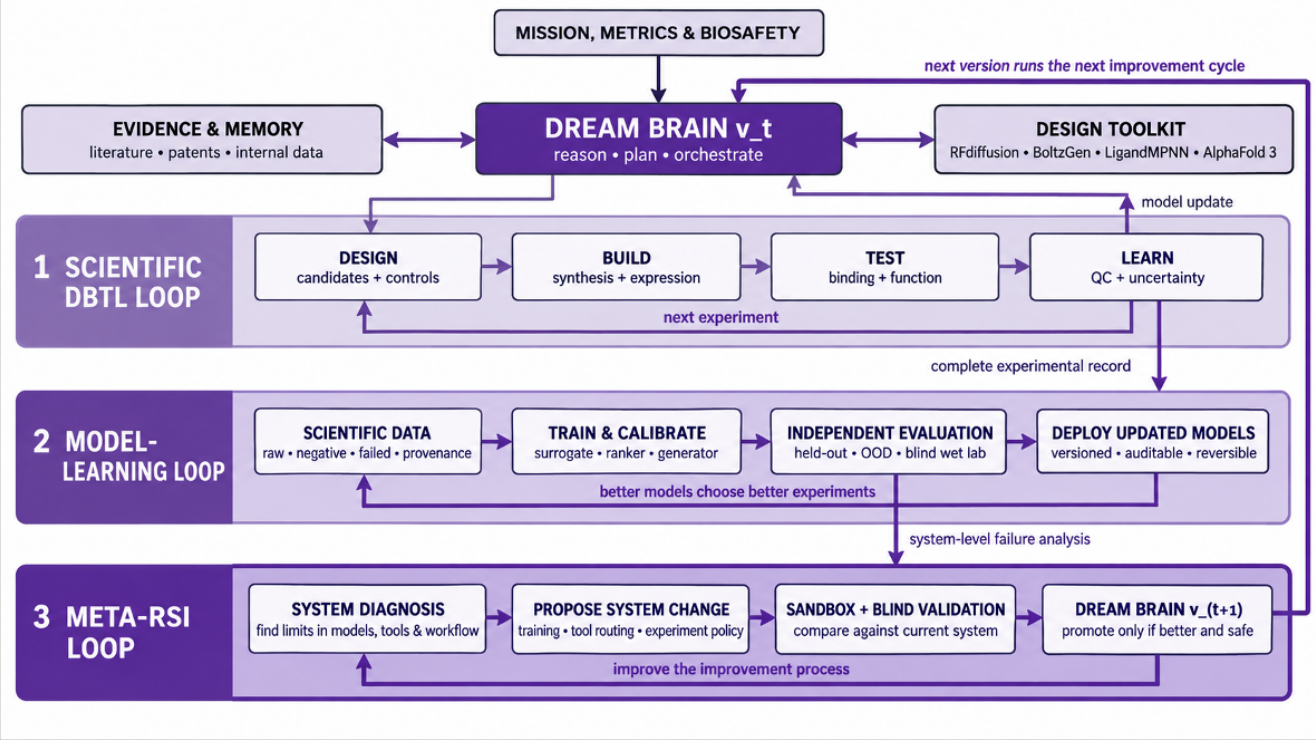


Figure 2. Three coupled loops in the proposed DREAM Brain architecture. Mission, metrics and biosafety define the operating context for system version v_t, supported by evidence, memory and a design toolkit. The scientific DBTL loop selects and interprets experiments; the model-learning loop uses qualified records to train, calibrate and independently evaluate model updates; the meta-RSI loop diagnoses system limitations and tests changes to the improvement process before promotion to v_{t+1}. Feedback connects the next experiment, model deployment and system revision. Named tools illustrate the proposed toolkit, not verified integrations. The diagram describes a proposed architecture, not demonstrated autonomous self-improvement.

system policies—and require separate validation. They describe a proposed development path, not demonstrated autonomous self-improvement.

2.1 Mission, evidence and tools define the operating context

An experimental campaign begins with a scientific objective, measurable acceptance criteria, a budget and constraints, including applicable biosafety requirements. "Improve binding" is incomplete without a target, comparator, assay and required operating conditions. DREAM Brain should translate the request into a campaign contract that the user and facility can review, recording the initial evidence, allowed design space, controls and conditions for advancing, revising or stopping.

A deployed system version, denoted v_t, reasons and plans within this contract, retrieves evidence and orchestrates task-specific services. Evidence and memory connect literature, patents where relevant, and internal experimental records to the current decision. Retrieved claims should retain locators, experimental context and a declared scope of applicability. An exact-substitution measurement, a homologous-position observation and a structural analogy remain distinct forms of support. Reference sequences, construct boundaries and numbering maps establish protein identity; unresolved mappings should prevent automatic reuse of variant-level claims.

The design toolkit supplies complementary proposals and assessments. Sequence models contribute evolutionary or generative priors, structure models assess geometric plausibility, and prior experiments constrain mechanistic hypotheses. Each output retains its method, version and uncertainty: co-folding confidence is not a binding free-energy difference, and sequence likelihood is not a measured phenotype. The architecture allows tools to be exchanged without changing the requirement to test their predictions experimentally. Representative foundations include evolutionary sequence models, large protein language models, structure prediction and learned sequence generation [29, 30, 31, 32, 33, 34, 35]. These methods motivate complementary toolkit components; their published capabilities do not establish performance on a DREAM campaign.

2.2 Loop 1: use experiments to choose the next experiment

The scientific DBTL loop converts a campaign objective into an interpretable experimental cycle. Each proposed cycle is represented by

$$\mathcal{C}_t = (G_t,\, X_t,\, P_t,\, A_t,\, R_t), \tag{1}$$

where G_t is the objective and constraint set, X_t the selected candidates and controls, P_t the prospective predictions, A_t the executable assay plan and R_t the interpretation and next-action rules. The cycle index and system version are tracked separately: multiple experimental cycles may use the same frozen system. This contract is an interface specification, not a claim of completed implementation.

During *design*, the planner selects promising candidates, informative contrasts and controls. Disagreement among models can motivate experiments that distinguish hypotheses rather than an average of incompatible scores. During *build* and *test*, synthesis, expression and functional measurements produce both observations and execution provenance. During *learn*, quality assessment precedes biological interpretation, and uncertainty informs whether to advance a candidate, add a control, repair the assay or stop. The immediate feedback is the next experimental decision; it need not involve retraining a model. Bayesian experimental design and Bayesian optimization provide formal approaches to selecting informative or promising experiments under resource constraints [36, 37, 38, 39]. Applying these approaches here would require an explicit objective, a suitable observation model and validation under the campaign's assay conditions.

Scientific orchestration manages campaign state, retrieval, model calls and review. Typed services handle operations such as identifier normalization, plate-map generation and data validation, while instrument drivers and validated manipulation skills retain timing-critical hardware control. Execution remains within approved workflows and operating limits. Protocol changes, approvals and human interventions must be recorded, consistent with DREAM's planned responsible-innovation and biosafety review process [5]. Under matched experimental budgets, the selection policy should be compared with expert selection, simple baselines and individual models; superiority is not established here.

The experimental budget includes controls and necessary repeats. Each panel member should have a declared role: testing a candidate, distinguishing explanations or qualifying the assay. An informative control may therefore take priority over a new sequence. When outcomes require a revised measurement or decision rule, the contract should preserve the original prediction and record the revision.

2.3 Loop 2: turn experimental records into better models

The model-learning loop begins with the complete experimental record rather than a collection of successful candidates alone. Raw measurements, valid negative outcomes, synthesis and assay failures, prospective predictions and provenance should remain linked in scientific memory. Their meanings must remain distinct: a failed assay does not supply a negative biological label. Scientific Process Data, described in Section 3, preserves the decisions and interpretations needed to reuse these records.

Qualified records can support training and calibration of surrogate models, rankers and generators. Candidate updates should pass independent evaluation using held-out campaigns, scaffold groups and temporal partitions, with out-of-distribution tests and blinded wet-lab validation where appropriate. Evaluation data must remain separate from the records used to select or train the update. Deployment should be versioned, auditable and reversible, so that subsequent experiments can be attributed to the models that informed them.

This loop returns improved models to DREAM Brain's planning process and toolkit. Its central hypothesis is that accumulated evidence enables better experimental choices across users, scaffolds and assays. A learning claim requires a frozen baseline, a defined update and improved performance on a subsequent independent task at a fixed experimental budget. Better fit to historical records alone is insufficient.

2.4 Loop 3: improve how the system improves

The meta-RSI loop addresses limitations that a model update alone may not resolve. System-level failure analysis should distinguish weaknesses in models, tool selection, experimental policy and workflow interfaces. For example, recurrent ambiguity may motivate a revised control-selection policy, while poor transfer may motivate a different training or calibration procedure. Each proposed change should identify the diagnosed limitation, the affected components and a prespecified evaluation criterion.

Proposed changes first enter a sandbox and undergo blinded comparison with the current system. Promotion from DREAM Brain v_t to v_{t+1} should require evidence of improvement on the declared criteria while satisfying the facility's safety and operating constraints. Model updates within a fixed workflow belong to the second loop; revisions to training procedures, tool-routing rules or experimental policies belong to this third loop. Changes to facility procedures, authority or the record schema require explicit review and versioning.

The promoted version then runs the next improvement cycle, closing the outer loop. This is a bounded proposal–evaluation–promotion process: the system may propose changes, but it does not grant itself additional authority or determine success solely through its own training reward. The three loops therefore pose three separate questions: did the experiment support a useful decision, did the model update improve future experiments, and did the system revision improve the process of learning? Each claim requires independent evidence under matched experimental budgets.

3 Scientific Process Data

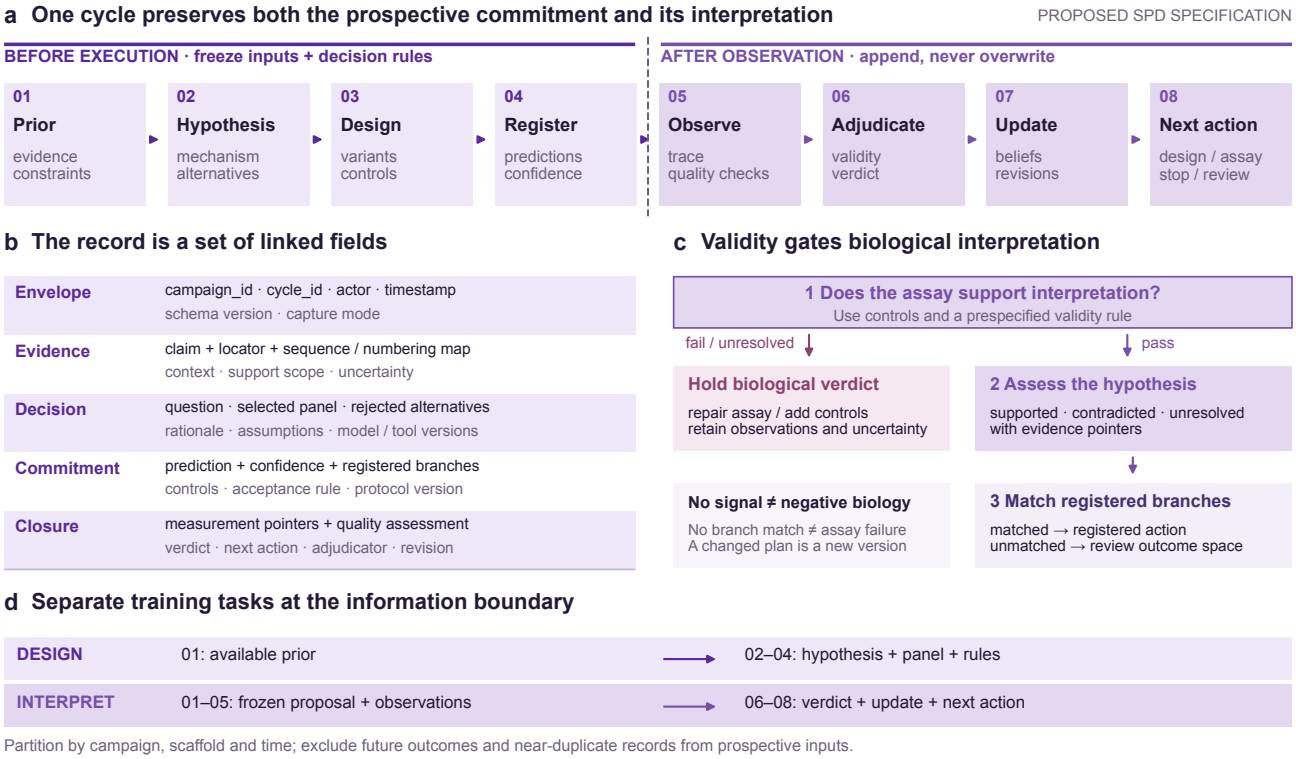


Figure 3. Scientific Process Data as an operational record. (a) Eight stages separate frozen prospective commitments from appended observations and interpretations. (b) Linked fields preserve identity, evidence, decisions, commitments and closure. (c) Assay validity gates biological interpretation; hypothesis assessment and branch coverage remain distinct. (d) Prospective design and retrospective interpretation use different inputs and targets. Campaign, scaffold and temporal partitions prevent future outcomes and near-duplicate records from leaking into evaluation.

An SPD record represents a decision at the point where its alternatives and justification can still be recovered. Its **envelope** identifies the campaign, cycle, actor, timestamp, schema version and capture mode (prospective, retrospective or agent-drafted). Its **decision block** contains the question, selected action, evidence pointers, assumptions, rejected alternatives and rationale. Its **closure block** links prospective predictions and their confidence to measurements, quality assessments, an adjudicated verdict and an authorized next action. Model-generated rationales are proposals to review, not privileged access to a scientist's or model's internal reasoning.

The distinction between experimental validity and scientific interpretation is essential. We propose separate fields for *assay validity* (pass, fail or unresolved), *hypothesis assessment* (supported, contradicted or unresolved) and *branch coverage* (matched or unmatched). An invalid assay cannot acquire a negative biological label simply because no signal was observed. An unexpected but valid observation can remain unmatched without being forced into a previously enumerated branch. The original predictions are retained when the outcome space is revised. This separation is consistent with the distinction between prospective specification and exploratory interpretation emphasized in work on preregistration [40]. Confidence assessment likewise requires a defined prediction target and an evaluation procedure; proper scoring rules and calibration methods provide relevant foundations [41, 42].

SPD can be implemented as an application-level extension over provenance infrastructure [8]. It should preserve raw data and link outward to laboratory information systems instead of replacing them. Capture should draw on existing design documents and instrument logs; expert edits, rejected suggestions and assay deviations are additional events. The time required to collect these records is an evaluation outcome. The current materials do not support a measured claim of thirty minutes per round or zero-cost capture.

3.1 From source artifacts to a linked cycle record

The supplied presentation provides a concrete example of the proposed capture workflow (Figures 4 and 5). Discussion transcripts preserve experimental context, while the design slide records the intended contrasts among C134R, C134F and C134E and the actions associated with four anticipated outcomes. These artifacts contribute different parts of a decision record: what was discussed, why the panel was selected and how possible observations would affect the next experiment.

The slide's mechanistic interpretations remain proposed decision rules, rather than established causal conclusions.

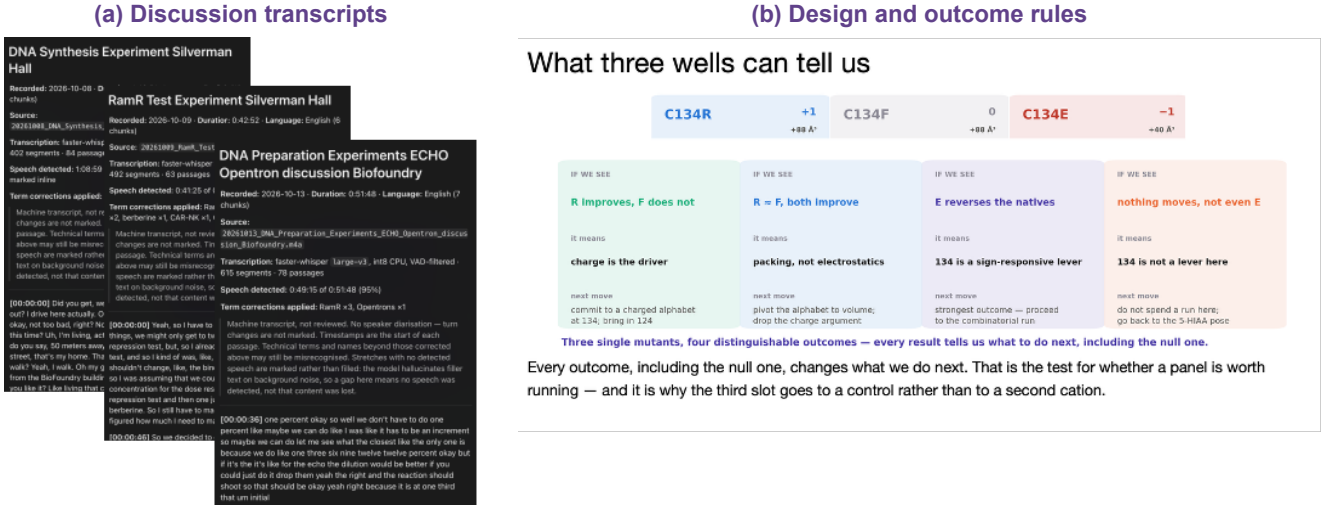


Figure 4. Source artifacts for Scientific Process Data. Supplied screenshots show (a) excerpts of laboratory discussion transcripts and (b) a RamR design slide connecting three substitutions to anticipated outcomes and next actions. The transcript headers identify machine transcription without review; technical-term corrections do not establish complete transcription accuracy. The slide preserves proposed interpretations, not validated mechanisms or an exhaustive outcome space. These are source-document excerpts, not raw instrument measurements.

The abridged compiled record associates a cycle identifier with a hypothesis, a stated confound, variant roles, conditional next actions, observation summaries and a quality-control flag. Its explicit charge–volume confound is valuable: the C134R–C134F comparison was intended to probe charge, but does not isolate charge from all other side-chain properties. Likewise, the recorded non-monotonic response at an intermediate dose motivates inspection; the suggested preparation or plate-column explanation remains a hypothesis about the anomaly.

Compiling these artifacts into reusable SPD requires field-level source locators, source and extraction versions, review status and temporal provenance. A transcript passage, slide region and measurement file should remain distinguishable even when they contribute to the same cycle. A field named `preregistered` requires a timestamped pre-execution source to support that status; otherwise the record should retain an explicit retrospective or unverified designation. Branch assessment also needs an explicit status so that missing assessment is not represented as a negative result. The presentation describes combining meeting audio, a proposal slide and a plate file, but the supplied compiled excerpt explicitly identifies plot-derived observations. The screenshots therefore do not establish a completed raw-file ingestion workflow.

The observation summaries also require reconciliation with Section 5. The compiled excerpt lists approximate 600-min best-to-uninduced ratios of 1.68, 1.16, 1.08 and 1.25 for WT, C134R, C134F and C134E, respectively. The manuscript's pilot summary instead reports fixed-dose F_{20}/F_0 ratios of 1.53, 1.01, 0.81 and 1.24 over a 9–10 h averaging window. These differ in endpoint definition and potentially in processing; they cannot be substituted or pooled without the underlying files, calibration and analysis provenance. The numbering correspondence shown in the compiled excerpt likewise remains a source assertion pending sequence verification.

Within the RSI architecture, such a record could inform the next experiment in the scientific loop. After review and task-specific partitioning, it could also supply examples for model learning. Recurrent compilation or interpretation errors could motivate a separately evaluated change to the system's procedures. The attachments illustrate the information needed for these connections; they do not demonstrate gains in any of the three loops.

Missing files and conflicting summaries require distinct review statuses. Corrections should identify the source and reason and append a linked revision. Retrieval should preserve construct identity, assay context, quality status and unresolved alternatives so that a reporter observation is not generalized into a binding claim. Even an unresolved record may help identify a missing control.

3.2 Learning tasks and temporal boundaries

Two learning tasks follow from the temporal structure. *Prospective design* uses the information available before execution to propose hypotheses, panels and outcome rules. *Retrospective interpretation* uses the frozen proposal plus observations to assess validity and choose the next action. Improvements on one do not establish improvements on the other. Both require data partitions that keep later rounds, near-duplicate records and closely related scaffolds from leaking into evaluation inputs. Dataset documentation and explicit analysis of leakage provide complementary practices for making these boundaries

```
// cycle RAMR-134-A · abridged
{
  "cycle_id": "RAMR-134-A",
  "system": "RamR / 5-HIAA cell-free biosensor",
  "position": { "mature": 134, "full_length": 135, "wt": "Cys" },

  "hypothesis": {
    "claim": "A positive charge at 134 improves the 5-HIAA response",
    "mechanism": "guanidinium reaches the 106 A^3 that 5-HIAA leaves empty vs berberine",
    "known_confound": "Cys->Arg adds charge and +87.8 A^3 together"
  },

  "arms": [
    { "id": "C134R", "role": "hypothesis", "charge": 1,  "d_vol_A3": 87.8 },
    { "id": "C134F", "role": "steric_twin", "charge": 0, "d_vol_A3": 88.3 },
    { "id": "C134E", "role": "sign_reversal + cross_validation", "charge": -1 },
    { "id": "WT",    "role": "assay_control", "charge": 0 }
  ],

  "preregistered": [
    { "if": "R improves, F does not", "then": "charge is the driver",
      "next": "charged alphabet at 134; bring in 124", "fired": false },
    { "if": "R ~ F, both improve", "then": "packing, not electrostatics",
      "next": "pivot alphabet to volume", "fired": false },
    { "if": "E reverses the natives", "then": "134 is a sign-responsive lever",
      "next": "proceed to combinatorial run", "fired": false },
    { "if": "nothing moves, not even E", "then": "134 is not a lever here",
      "next": "back to the 5-HIAA pose", "fired": false }
  ],

  "observation": {
    "readout": "uM FITC, 0-600 min", "ligand_uM": [0, 1.28, 3.2, 8, 20],
    "endpoint_600min": {
      "WT":    { "uninduced": 0.345, "best": 0.580, "fold": 1.68 },
      "C134R": { "uninduced": 2.680, "best": 3.100, "fold": 1.16 },
      "C134F": { "uninduced": 0.072, "best": 0.078, "fold": 1.08 },
      "C134E": { "uninduced": 0.340, "best": 0.425, "fold": 1.25 }
    },
    "provenance": "read from plotted traces; replace with raw plate file",
    "qc_flags": [
      { "code": "NON_MONOTONIC_SERIES", "at_uM": 8,
        "seen_in": ["WT", "C134E", "C134R"],
        "note": "fails between two working concentrations; prep or plate-column artefact" }
```

Figure 5. An abridged compiled RamR cycle record supplied in the presentation. The screenshot illustrates links among hypothesis, panel, outcome rules, observations and quality flags. Its provenance field states that observation values were read from plotted traces and should be replaced with the raw plate file. The field named `preregistered` does not independently establish registration timing, and `fired: false` does not distinguish an unassessed branch from an assessed non-match. This artifact illustrates a representation; it does not validate an automated compiler or establish improved sensing.

auditable [43, 44].

4 Lab Automation

A programmable laboratory must handle both instrument-accessible operations and the physical steps between instruments. Drivers provide a consistent interface to supported protocol actions. Manipulation skills can address tasks such as plate transfer and lid handling where the workcell supports them. For each skill, the relevant unit is a verified state transition: the named object starts in a known state, an action is attempted, and the resulting state is observed. Success cannot be inferred solely from the absence of an API error. Published work on automated synthesis and executable chemical descriptions provides relevant precedents for translating experimental procedures into machine-operable workflows [45, 46]. For protein production, cell-free systems provide an experimental foundation for prototyping and characterization, while retaining requirements for condition-specific qualification [47, 48, 49].

The supplied simulation asset provides a spatial representation for developing these interfaces. Extending it into an execution gate requires explicit state variables, including labware identity, location, occupancy, liquid volume and relevant instrument status. Preflight checks should reject impossible or inconsistent actions within the model's scope. Geometry checks, volume accounting and scheduling checks require different validation sets; visual fidelity does not establish their correctness. Hardware observations must also be reconciled with expected state before subsequent actions are authorized.

The execution receipt should distinguish the requested action, device response and observed state, with declared verification

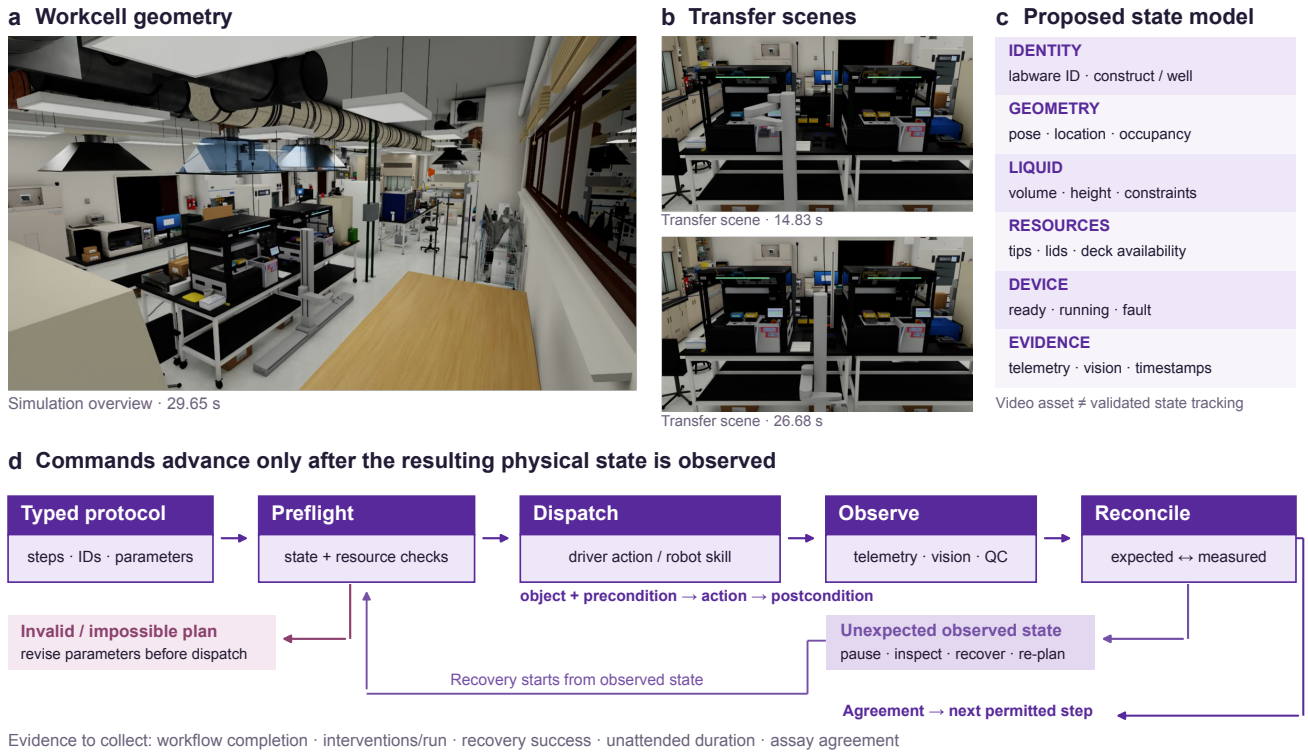


Figure 6. From a visual twin to a verified execution interface. (a,b) Unaltered frames from the supplied simulation video show the workcell and two transfer configurations. (c) A proposed state model identifies the information needed to validate execution; the video does not establish that these variables are tracked. (d) A typed protocol proceeds through preflight, dispatch, observation and reconciliation. Invalid plans require revision before dispatch; observed mismatches require recovery from the measured state. Hardware synchronization, recovery performance and assay agreement remain to be validated.

methods and explicit inferred properties. Command completion alone does not verify plate identity or delivered volume. Recovery must account for partially completed actions: an uncertain transfer should pause dependent steps until observations or human inspection resolve the state. Resumed workflows should retain deviations for downstream assessment of measurement comparability. These are proposed requirements, not capabilities established by the simulation.

Adaptive execution is a separate capability to demonstrate. If a readout is outside the assay's useful range, the system may propose a revised measurement plan, subject to recorded constraints and review. That proposal, its approval and the resulting observations should remain linked to the original cycle. The proposed evaluation therefore measures completion, intervention burden, recovery from faults and assay agreement with reference execution. Neither unattended runtime nor manipulation success is quantified by the current video.

5 Case Study: RamR Design

RamR, an allosteric transcriptional repressor, has been engineered into small-molecule biosensors [9]. Structural studies of RamR and broader work on the TetR family provide mechanistic context for ligand recognition and transcriptional regulation [50, 51]; transcription-factor-based biosensors also offer a broader framework for engineering and evaluating sensing behavior [52]. These precedents do not establish the mechanism of the substitutions in the supplied pilot. The supplied pilot compares three substitutions at one pocket position with a wild-type control. The panel was designed by humans to probe whether contrasting side-chain properties could help interrogate recognition of the target metabolite. The record describes four outcome branches prepared before measurement, but the timestamped registration and raw experimental files are not included in the current materials. We retain this as a preliminary decision case, not an autonomous-system benchmark.

C134R exhibits a basal signal about 7.87 times WT and an endpoint ratio near one. C134F has a basal signal about 0.21 times WT and a ratio below one. C134E remains close to WT in basal signal. These observations demonstrate substantial differences in reporter output under the reported conditions. They do not isolate a molecular mechanism: altered repression, expression, folding, sample preparation or reporter behavior remain possible contributors. In particular, similar side-chain volumes do not make an arginine–phenylalanine comparison a clean causal isolation of charge.

The draft also reports non-monotonic intermediate doses, variable replicates and the absence of a known-ligand positive control. Without the underlying traces and a qualified positive response, a ratio above one cannot establish improved sens-

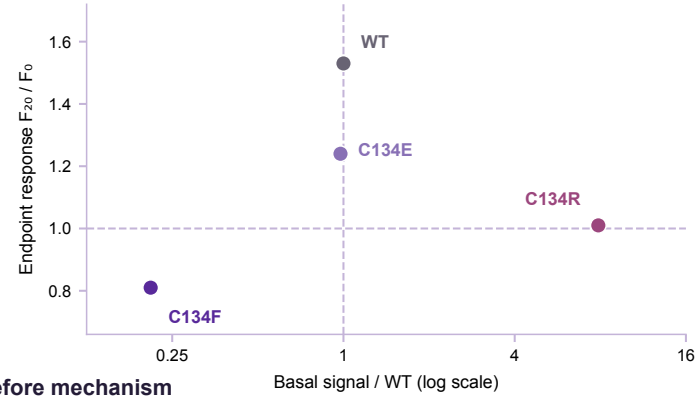
a Human-designed RamR pilot

Position 134 panel · 3 variants + WT

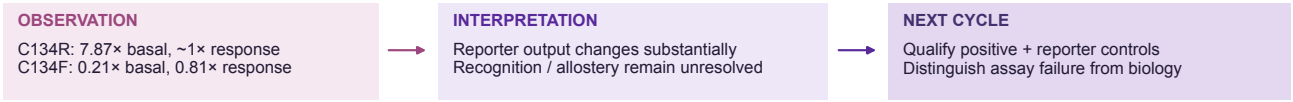
Variant	Basal g_0	g_0 / WT	F_{20} / F_0
WT	4,334	1.00×	1.53
C134R	34,120	7.87×	1.01
C134F	910	0.21×	0.81
C134E	4,231	0.98×	1.24

Reported: n = 3; mean signal over 9–10 h
Endpoint: 20 µM / 0 µM; raw replicates absent
No uncertainty estimates or curves reconstructed

b Basal shift and endpoint response are different axes



c The next experiment must resolve assay validity before mechanism



Summary data only. Dashed lines denote ratio = 1, not assay-validity or significance thresholds. No improved sensing is established.

Figure 7. The RamR pilot separates basal output from endpoint response. (a) Reported basal means and endpoint ratios for WT and three variants; basal ratios are recomputed relative to WT. (b) The same four profiles plotted jointly, with a logarithmic basal axis. Dashed reference lines indicate a ratio of one; they are not significance or assay-validity thresholds. The source reports a 9–10 h averaging window and $n = 3$, but supplies no raw replicates or uncertainty estimates. (c) Substantial reporter shifts motivate assay qualification before mechanistic interpretation or panel expansion.

ing. Repeated significance tests across correlated timepoints would not supply independent experimental replication. The defensible interpretation is that target-recognition and allosteric hypotheses remain unresolved.

A prospective follow-up should qualify the assay with known-ligand and reporter controls before comparing variants under matched conditions and verified construct identities. These controls remain proposed. The analysis should report basal signals and ligand-dependent changes separately, prespecify dose comparisons, averaging windows, replicate definitions and exclusions, and retain raw traces. Failed positive controls leave the biological verdict unresolved; qualified negative responses can inform design within the tested conditions. Neither outcome alone establishes a molecular mechanism.

The useful system lesson is a change in the next decision. A branch set organized only around "improvement" can fail to represent a shifted baseline, an invalid assay or an unexpected phenotype. The next cycle should first distinguish these possibilities with appropriate controls and a prespecified analysis. That decision can be recorded and evaluated even when the round produces no validated improved protein. The pilot motivates SPD; it does not establish that SPD training improves a model.

6 Additional Case Studies

The rationale for a shared architecture is that scientific objectives change while core obligations remain: define a measurable claim, select informative experiments, preserve execution context and determine what the observation licenses next. Lanmodulin and PduA are therefore prospective tests of transfer rather than additional completed demonstrations.

Lanmodulin: distinguish binding from selectivity. The discovery of lanmodulin established a protein platform for selective lanthanide recognition [53]. Lanmodulin-family proteins provide a platform for rare-earth recognition and separation; published work demonstrates that homolog-dependent and metal-dependent behavior can alter discrimination [10, 11]. A DREAM campaign should specify the target and comparator ions, assay conditions, reference binder and selectivity endpoint. Improving affinity alone need not improve separation. Evaluation should hold out homolog backgrounds and compare candidate-selection policies at equal experimental cost. No variant counts, selectivity gains or ranking improvements are reported here for this campaign.

PduA: distinguish assembly from function. PduA is a hexameric shell protein whose pore properties contribute to molecular transport in bacterial microcompartments [12]. Broader structural and functional studies of bacterial microcompartments provide context for distinguishing shell assembly, permeability and compartment function [54, 55]. The proposed metal-associated scaffold campaign requires at least two readouts: whether the intended assembly is retained and whether the desired function is present. A two-dimensional decision table can represent successful assembly without functional improvement, functional signal in an aberrant assembly, success on both dimensions, or failure on both. Assembly is an experimentally

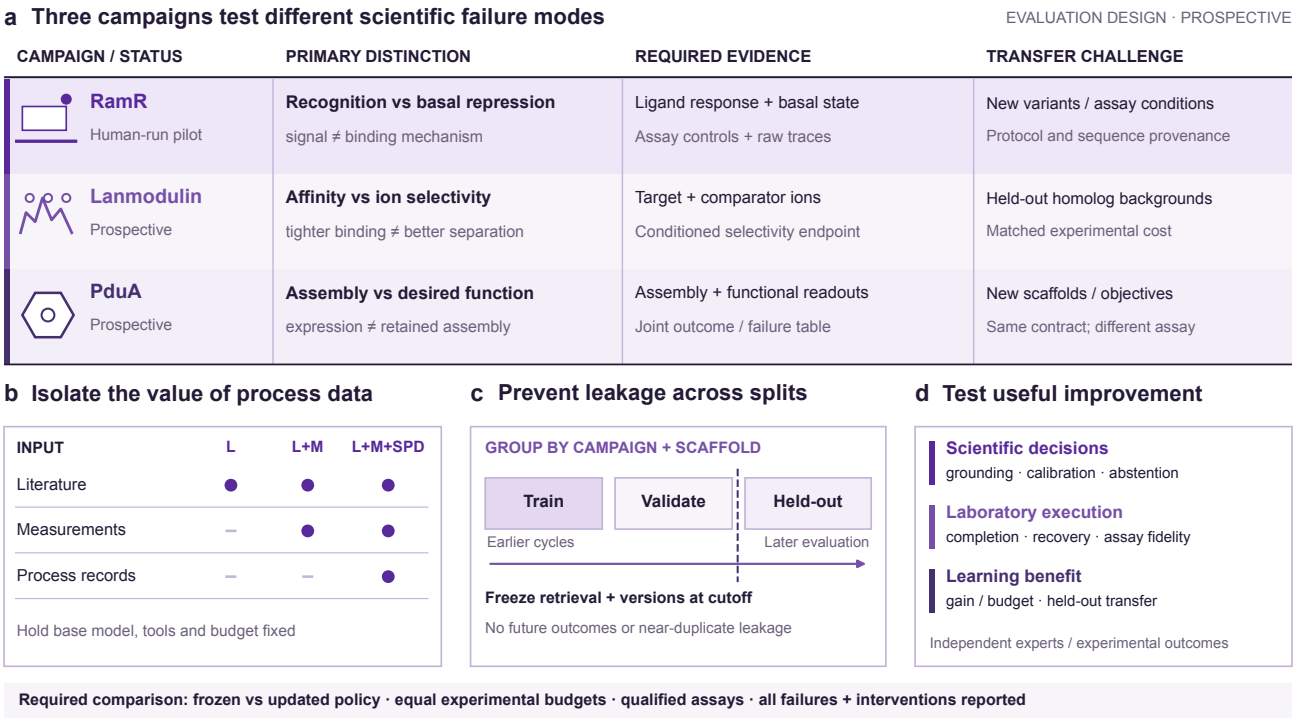


Figure 8. Scientific breadth and a falsifiable learning experiment. (a) The campaigns distinguish recognition from repression, affinity from selectivity, and assembly from function. RamR has preliminary summary data; Lanmodulin and PduA remain prospective. Small molecular symbols are schematic. (b) An input ablation compares literature only (L), literature plus measurements (L+M), and both plus process records (L+M+SPD). Filled dots indicate included input types, not measured performance. (c) Grouped and temporal partitions separate training from evaluation. (d) Decisions, execution and learning require distinct independent endpoints under matched budgets.

defined property; expression alone is not an adequate substitute.

For Lanmodulin, a useful prospective panel should distinguish a general increase in binding from a change in discrimination between the designated ions. Target and comparator measurements should therefore remain linked to the same construct and experimental conditions. A candidate that improves both measurements similarly would require a different interpretation from one that preferentially improves the target endpoint. If the reference binder or assay controls fail, the next action should address measurement validity before extending the sequence panel. These conditional decisions should be recorded before results are available, with numerical acceptance criteria chosen during assay qualification.

For PduA, the joint readouts should determine which follow-up is informative. Retained assembly without the desired functional change would motivate reconsidering the functional design while preserving assembly as a constraint. A functional signal accompanied by altered assembly would instead require testing whether the signal depends on the intended scaffold state. In either case, an additional measurement may be more useful than another mutation. These are proposed decision branches, not outcomes observed in the present materials.

Transfer between campaigns should initially be tested at the level of these decision practices: selecting controls, identifying confounded comparisons and requesting evidence that can resolve them. A RamR reporter threshold should not become a Lanmodulin selectivity criterion, nor should a PduA assembly verdict become a general sequence-quality label. Reusable memory must carry the conditions under which each conclusion applies.

Across these campaigns, the platform should reuse identities, provenance, decision records and interfaces while allowing the assays and interpretation rules to differ. The broader hypothesis is that consistent experimental contracts make learning transferable. Establishing that hypothesis requires completed cycles across backgrounds and objectives; it cannot be inferred from drawing the same workflow for three projects.

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